

Lab 1 Analysis — WalkMates

Course: Principles & Practices in Software Testing (DV033G)

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Part A — Fundamentals (M1)

Activity 1.1 — Quality-Attribute Analysis (ISO/IEC 25010)

FR-1.1 Registration

- Functional Suitability: System should only allow valid input and reject invalid
- Interaction capability: Seeker need to understand why a registration was rejected

FR-3.1 Types and base rate

- Functional Suitability: System should only allow one type
- Flexibility: Can a new listing type be added without scattered changes

FR-5.1 Match explanation

- Functional Suitability: The AI Feature should give the seeker relevant matches
- Maintainability: Can LLM clients be changed and tested efficiently

Testable Quality Requirement

FR1.1 Registration: For Functional Suitability, I would test whether the system accepts display names containing 2-40 characters when registering a Seeker.

Activity 1.2

Trace the chain: what was the likely human error, the fault in the code, and the failure the user saw?

Human Error: The developer used the wrong operator or misunderstood the requirements. Fault: Code does not reject a second booking when a new Seeker already has one active booking. Failure: The new Seeker has two active bookings even if the tier only allow for one booking.

Find the responsible code (hint: the FR-4.4 rule-2 check). Describe the fault precisely.

Responsible code: `if (seekerActive > seeker.getMaxConcurrentBookings())` . This is a boundary issue since the FR-4.4 rule-2 states that the The Seeker's active Bookings < the trust-tier max . The trust-tier maximum for active bookings for a New Seeker is 1. However, the code checks whether the seeker's current active bookings are greater than `getMaxConcurrentBookings` before rejecting. A New Seeker with 1 active booking is therefore still allowed to create a second booking. The system only rejects the booking when the number of current active bookings is greater than 1.

Which test level (unit / integration / system) should have caught this, and what kind of test (which technique)?

A Unit Test should've caught this defect. This would've been caught with a simple Arrange-Act-Assert test structure using boundary value analysis.

Part B — Specification-Based Design (M2)

Activity 2.1 — Equivalence Partitioning (Seeker Unit)

Derived from FR-1.1 and FR-1.3 for the Seeker class.

Input Field	Partition ID	Description	Representative Input	Expected Outcome
Email	EP-EM-VAL	Valid format, length ≤ 254	test@example.com	Accepted
	EP-EM-INV-1	Invalid: missing @	testexample.com	Rejected (IllegalArgumentExceptionException)
	EP-EM-INV-2	Invalid: empty local part	@example.com	Rejected (IllegalArgumentExceptionException)
	EP-EM-INV-3	Invalid: missing . in domain	test@example	Rejected (IllegalArgumentExceptionException)
	EP-EM-INV-4	Invalid: length > 254 chars	"a".repeat(246) + "@example.com"	Rejected (IllegalArgumentExceptionException)
	EP-EM-INV-5	Invalid: multiple @ characters	sam@@example.com	Rejected (IllegalArgumentExceptionException)
Display Name	EP-DN-VAL	Valid: 2-40 chars, letters/spaces/hyphens/apostrophes	O'Connor-Smith Jr	Accepted
	EP-DN-INV-1	Invalid: too short (< 2 chars)	A	Rejected (IllegalArgumentExceptionException)
	EP-DN-INV-2	Invalid: too long (> 40 chars)	"Name".repeat(11) (44 chars)	Rejected (IllegalArgumentExceptionException)
	EP-DN-INV-3	Invalid: forbidden characters	Sam123!	Rejected (IllegalArgumentExceptionException)
Phone Number	EP-PH-VAL-1	Valid Swedish format (07 + 8 digits)	0701234567	Accepted
	EP-PH-VAL-2	Valid Swedish international format (+467 + 8 digits)	+46712345678	Accepted

Input Field	Partition ID	Description	Representative Input	Expected Outcome
	EP-PH-INV-1	Invalid format (does not start 07/+467)	0812345678	Rejected (IllegalArgumentExceptionException)
	EP-PH-INV-2	Invalid length (too short)	07012345	Rejected (IllegalArgumentExceptionException)
	EP-PH-INV-3	Invalid length (too long)	07012345678	Rejected (IllegalArgumentExceptionException)
Wallet Top-Up	EP-TU-VAL	Valid single top-up amount [10.00, 5000.00]	100.00	Accepted
	EP-TU-INV-1	Invalid: top-up amount below 10.00 SEK	9.99	Rejected (IllegalArgumentExceptionException)
	EP-TU-INV-2	Invalid: top-up amount above 5,000.00 SEK	5000.01	Rejected (IllegalArgumentExceptionException)

Activity 2.2 — Boundary Value Analysis (Wallet Top-Up)

Derived from FR-1.3 for Seeker.addFunds(double amount) starting with a balance of 0.00 (unless otherwise noted).

Boundary Base	Test Case ID	Test Value	Expected Outcome	Justification / Note
Min Top-up (10.00)	BVA-MIN-BELOW	9.99	Rejected (IllegalArgumentExceptionException)	Just below the minimum limit
	BVA-MIN-AT	10.00	Accepted (Balance: 10.00)	Exactly at the minimum limit
	BVA-MIN-ABOVE	10.01	Accepted (Balance: 10.01)	Just above the minimum limit
Max Single (5000.00)	BVA-MAX-BELOW	4999.99	Accepted (Balance: 4999.99)	Just below the single transaction maximum
	BVA-MAX-AT	5000.00	Accepted (Balance: 5000.00)	Exactly at the single maximum (worked example)
	BVA-MAX-ABOVE	5000.01	Rejected (IllegalArgumentExceptionException)	Just above the single maximum limit

Boundary Base	Test Case ID	Test Value	Expected Outcome	Justification / Note
Max Balance (20000.00)	BVA-BAL-BELOW	Top up 1999.99	Accepted (Balance: 19999.99)	Balance just below 20000.00 (start balance: 18000.00)
	BVA-BAL-AT	Top up 2000.00	Accepted (Balance: 20000.00)	Balance exactly at 20000.00 (start balance: 18000.00)
	BVA-BAL-ABOVE	Top up 2000.01	Rejected (IllegalArgumentException)	Balance would exceed 20000.00 (start balance: 18000.00)

Activity 2.3 — Decision Table (Trust Tier → Limits)

Derived from FR-1.2 for testing the correlation between a Seeker 's trust tier and their maximum active bookings and platform fee.

Rule / Trust Tier	Max Concurrent Bookings	Platform Fee	Test Representative
Rule 1: NEW	1	15% (0.15)	TrustTier.NEW
Rule 2: VERIFIED	3	12% (0.12)	TrustTier.VERIFIED
Rule 3: TRUSTED	5	8% (0.08)	TrustTier.TRUSTED
Rule 4: PRO_SITTER	10	5% (0.05)	TrustTier.PRO_SITTER